



mnirs

v0.4.2.2

Muscle Near-Infrared Spectroscopy Processing and Analysis

Read, process, and analyse data from muscle near-infrared spectroscopy (mNIRS) devices. Import raw data from .csv or .xls(x) files and return a time-series data frame with metadata. Includes standardised methods for cleaning, filtering, and pre-processing mNIRS data for subsequent analysis, with a custom plot theme and colour palette.

Workflow

```
└── read_mnirs()
|   ├── create_mnirs_data()
|   └── example_mnirs()
└── resample_mnirs()
    ├── replace_mnirs()
    |   ├── replace_invalid()
    |   ├── replace_outliers()
    |   └── replace_missing()
    └── filter_mnirs()
        ├── filter_ma()
        └── filter_butter()
└── shift_mnirs()
    ├── rescale_mnirs()
    └── plot.mnirs()
        ├── theme_mnirs()
        ├── palette_mnirs()
        ├── scale_colour_mnirs()
        ├── scale_fill_mnirs()
        ├── format_hmmss()
        └── breaks_timespan()
└── extract_intervals()
```

Read

`read_mnirs()` — Import mnirs data from .csv or .xls(x)

Package-level functions

`example_mnirs()` — Paths to an included example file

`create_mnirs_data()` — Add “mnirs” class metadata to data

Resample

`resample_mnirs()` — Resample to a regular time grid

Clean

`replace_mnirs()` — Replace outliers, invalid, and missing data

Vector-level functions

`replace_outliers()` — Replace local outliers

`replace_invalid()` — Replace specified invalid values

`replace_missing()` — Interpolate over NA values

Filter

`filter_mnirs()` — Apply digital filter to mNIRS channels

Vector-level functions

`filter_butter()` — Apply a Butterworth digital filter

`filter_ma()` — Apply centred moving average smoothing

Transform

`shift_mnirs()` — Shift channels to a reference value

`rescale_mnirs()` — Rescale channels data range

Plot

`plot()` — Plot a data frame of class “mnirs”

`theme_mnirs()` — ggplot2 theme for mnirs plots

`palette_mnirs()` — Access the mnirs colour palette

`scale_colour_mnirs()` — Colour scale with mnirs palette

`scale_fill_mnirs()` — Fill scale with mnirs palette

`format_hmmss()` — Plot axis in time units

`breaks_timespan()` — Plot axis with time breaks

Extract

`extract_intervals()` — Detect events and extract intervals